FACULTY OF COMPUTING & INFORMATION TECHNOLOGY

BSc (Hons) in Computer Science Handbook

Code: CS-PROG-2026
Academic Year: 2026/2027

1. Programme Overview & Architecture

The Bachelor of Science (Honours) in Computer Science degree is a four-year full-time undergraduate programme designed to produce competent, research-oriented, and industry-ready computer professionals. Students undergo rigorous theoretical training in algorithmic complexity, distributed systems, mathematical foundations, and applied machine learning.

2. Year 3 Core & Elective Module Specifications

Students in Year 3 must complete 60 credits across Semester 1 and Semester 2. Below is the detailed structural breakdown of all core modules offered in the academic year.

Code	Module Title	Credits	Prerequisites	Assessment Breakdown
CS301.3	Advanced Algorithms & Complexity	4 Credits	CS106.3 (Data Structures)	60% Exam, 40% Continuous Assessment
CS303.3	Computational Theory & Automata	4 Credits	MA101.3 (Discrete Math)	70% Exam, 30% Coursework
CS305.3	Compiler Design & Construction	4 Credits	CS102.3, CS104.3	50% Project, 50% Final Exam
CN302.3	Wireless & Distributed Networks	3 Credits	CN101.3 (Data Networks)	40% Lab Practical, 60% Exam
SE308.3	Software Architecture & Quality Assurance	3 Credits	SE101.3 (OOP Java)	50% Group Project, 50% Exam
CS306.3	Information Assurance & Security	3 Credits	CN101.3, CS105.3	40% Security Audit, 60% Exam
CS302.3	Applied Cryptography	3 Credits	MA301.3 (Advanced Math)	50% Cryptanalysis Lab, 50% Exam

3. Key Module Descriptions

CS301.3 Advanced Algorithms & Complexity: Covers NP-completeness, randomized algorithms, approximation algorithms, and dynamic programming optimization techniques. Students implement custom Graph algorithms in C++ or Rust.

CS306.3 Information Assurance & Security: Focuses on network defence, threat modeling, zero-trust architectures, security compliance standards (ISO 27001), and vulnerability assessments using open-source audit tools.